SmartRoute AI — Market Requirements Document (MRD)

The Business Case, Market Need & Real-World Problem — Explaining SmartRoute AI in Simple English

DOCUMENT: MRD-v1.0
AUDIENCE: Judges & Stakeholders
STATUS: Final Hackathon Pitch

The 30-Second Elevator Pitch for Judges:
Most navigation apps blindly suggest the shortest route, even if it leads straight into heavy traffic gridlock. **SmartRoute AI (Route Wise)** is a smart API and live dashboard that compares multiple routes, penalizes severe traffic delays, and picks the truly best route. It explains **why** it chose the route in plain words (e.g. *'Saves 16 mins of traffic delay for just 2 km extra'*), saving time, fuel, and driver frustration for delivery fleets and mobility apps.

1. Real-World Story: Why Traditional GPS Fails Drivers

The Scenario (Meet Alex the Courier): It is 5:15 PM during evening rush hour. Alex has an urgent package to deliver from Downtown to the Airport Logistics Hub.

- **What Traditional GPS Does:** It recommends *Route A (Direct City Avenue)* because it is only **10.0 km**. However, City Avenue has an unexpected bottleneck with **18 minutes of bumper-to-bumper traffic delay**. Total travel time: **42 minutes**. Alex arrives late, the customer complains, and fuel is burned idling in traffic.
- **What SmartRoute AI Does:** SmartRoute AI analyzes 3 alternatives simultaneously. It recognizes that *Route B (Outer Ring Road)* is **12.5 km** (2.5 km longer), but has almost zero congestion (only **2 minutes of delay**). Total travel time: **26 minutes**. SmartRoute AI recommends Route B, explaining: *'Fastest path — Avoids 16 minutes of traffic jam'*. Alex arrives 16 minutes early!

Navigation Strategy	Distance	Traffic Delay	Total Time	Driver & Customer Outcome
Standard Navigation App (Picks shortest distance)	10.0 km (Shortest)	**+18 mins** (Heavy gridlock)	**42 mins**	Late delivery, high stress, fuel wasted idling in traffic jam.
SmartRoute AI (Route Wise) (Multi-criteria traffic scoring)	12.5 km (+2.5 km)	**+2 mins** (Smooth flowing)	**26 mins (16m faster!)**	**On-time delivery, clear explanation, fuel-efficient constant speed.**

2. The Big Market Problem We Solve

1. **Opaque 'Black-Box' Algorithms:** Current mapping APIs provide a single route without explaining why it was picked, leaving dispatchers blind.
2. **Inability to Customize Priorities:** A courier company cares deeply about avoiding red-light gridlock, while an EV driver cares about shortest distance. Standard APIs give everyone the exact same default.
3. **Demo & Network Fragility:** Traditional routing APIs break when offline or over quota. Developers need high-reliability fail-safes.

3. Market Size: Multi-Billion Dollar Opportunity

TAM: $14.2 Billion Global GIS, Navigation & Fleet Logistics API Market	**SAM: $2.8 Billion** Urban Last-Mile Delivery & Courier Routing Software	**SOM: $350 Million** Independent Logistics SaaS, Startups & Developers

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4. Target Audience, Competition & Revenue Model

4.1 Who Needs This? (Target Customer Segments)

- **On-Demand Couriers & Food Delivery Platforms (e.g., DoorDash, UberEats, Local Couriers):** Need to avoid traffic jams to ensure meals stay hot and deliveries meet promised ETAs.
- **Fleet & Logistics Dispatch Managers:** Managing 10 to 100 delivery vans. A 15-minute delay per driver compounds into thousands of lost dollars and overtime costs every week.
- **Independent Developers & Tech Startups:** Need a clean, predictable REST API with built-in API keys and guaranteed offline demo fallback for rapid integration.

4.2 Why SmartRoute AI Wins Against Big Tech APIs

Feature / Capability	Google Maps API	Mapbox Navigation	SmartRoute AI (Route Wise)
Configurable Scoring Weights	**No** (Fixed default algorithm)	**Partial** (Limited vehicle profiles)	**Yes — Customize Time, Delay Penalty & Distance**
Transparent 'Why' Rationale	**No** (Opaque black-box route)	**No** (No explanation given)	**Yes — Clear tags like 'Avoids 16m gridlock'**
Traffic Severity Badges	**Partial** (Visual map lines only)	**Partial** (Generic speed colors)	**Yes — Green / Yellow / Red delay classifier**
Offline Demo Fail-Safe	**No** (Fails if internet drops)	**No** (Requires paid offline packs)	**Yes — High-fidelity local offline dictionary**
Developer Dashboard & Keys	**Partial** (Complex cloud billing)	**Partial** (Cluttered UI console)	**Yes — Instant modal with cURL & SHA-256 keys**

4.3 How We Monetize (Simple Tiered Model)

Tier	Price	Included Quota & Capabilities	Target Customer
Free Developer	**$0 / mo**	1,000 queries/month, 60 req/min rate limit, full dashboard access, offline mock mode.	Hackathons, student projects, prototyping.
Growth Fleet	**$0.003 / query**	Up to 250,000 queries/month, custom scoring weight presets, 60-second live refresh API.	Courier startups, local delivery fleets.
Enterprise	**Custom SLA**	Unlimited queries, multi-stop TSP route optimization, dedicated container hosting.	Logistics corporations, mobility providers.

4.4 Top 3 Takeaways to Explain to Judges

1. Not Just Another Map: We don't reinvent map tiles; we solve the intelligence layer that decides which route is best based on traffic delays.

2. Tangible Business ROI: Saving 15 minutes of traffic delay per vehicle directly reduces fuel costs, missed deliveries, and driver burnout.

3. Production-Ready Tech: Built with real-time TomTom traffic telemetry, instant offline fallbacks, secure SHA-256 keys, and 22 automated tests.